Statistical Field Theory in Soft Matter: From Microscopic Ising-Yukawa Model to Local Critical Phenomena

Jeong Kangeun

April 24, 2026

1 Ising Model with Yukawa Interactions: A Rigorous Field-Theoretic Approach

In this section, we will walk through the exact mathematical steps required to translate a microscopic, discrete lattice model into a macroscopic, continuous field theory. We will use a magnetic system as our example: an Ising-like model where spins ($s_i = \pm 1$) interact via a non-local, ferromagnetic Yukawa potential V_Y in the presence of an external magnetic field h.

Our goal is to make every mathematical transition—especially the jump from discrete points to a continuous space—crystal clear.

1.1 Step 1: The Mathematical Bridge (Continuous Spin Density)

In traditional lattice models, we describe interactions using discrete sums over lattice sites i and j. However, field theory operates in a continuous space ($\mathbf{r}$). To rigorously connect these two worlds, we introduce the **microscopic spin density** $\rho(\mathbf{r})$.

Using the Dirac delta function $\delta(\mathbf{r}-\mathbf{r}_i)$, which acts as a mathematical "spike" that is exactly zero everywhere except at the position $\mathbf{r}_i$, we can write the spin density as a continuous spatial function:

$$\rho(\mathbf{r}) = \sum_i s_i \delta(\mathbf{r} - \mathbf{r}_i) \tag{1}$$

Similarly, we represent the external magnetic field as a continuous function $h(\mathbf{r})$.

By using this continuous density, we can rewrite the entire Hamiltonian $\mathcal{H}$ using spatial integrals instead of discrete sums. The interaction between two spins at different locations becomes an interaction between two density points:

$$\mathcal{H} = -\frac{1}{2}\int d\mathbf{r}d\mathbf{r}'\rho(\mathbf{r})V_Y(\mathbf{r}-\mathbf{r}')\rho(\mathbf{r}') - \int d\mathbf{r}h(\mathbf{r})\rho(\mathbf{r}) \tag{2}$$

Note: When we replace the discrete sum $\sum_{i\neq j}$ with continuous integrals, we accidentally include the case where $\mathbf{r} = \mathbf{r}'$ (a spin interacting with itself). This generates an infinite "self-energy". However, because Ising spins have a fixed magnitude ($s_i^2 = 1$), this self-energy is merely a constant additive value. In statistical mechanics, adding a constant to the energy does not change the physical behavior or phase transitions, so we can safely ignore it.

The grand partition function Z, which is the sum of Boltzmann weights over all possible spin configurations $\{s_i\}$, is now written purely in terms of continuous integrals:

$$Z = \sum_{\{s_i\}} \exp\left(\frac{\beta}{2}\int d\mathbf{r}d\mathbf{r}'\rho(\mathbf{r})V_Y(\mathbf{r}-\mathbf{r}')\rho(\mathbf{r}') + \beta\int d\mathbf{r}h(\mathbf{r})\rho(\mathbf{r})\right) \tag{3}$$

1.2 Step 2: Decoupling via Hubbard-Stratonovich Transformation

The partition function Z contains a quadratic term ($\rho \times \rho$) in the exponential, which is mathematically notoriously difficult to calculate because the spins are entangled.

To "decouple" them, we use the **Hubbard-Stratonovich (HS) transformation**. Think of this as a mathematical trick: instead of particles interacting directly with each other, we imagine they interact independently with a newly introduced "auxiliary field" $\psi(\mathbf{r})$.

Since ferromagnetic spins attract, the interaction energy lowers the total energy, leading to a positive sign in the exponential. This allows us to use a **real** continuous field $\psi(\mathbf{r})$. Applying the HS identity using the inverse Yukawa operator $V_Y^{-1} \propto (-\nabla^2 + \kappa^2)$, we get:

$$\exp\left(\frac{\beta}{2}\int \rho V_Y \rho\right) \propto \int \mathcal{D}\psi \exp\left(-\frac{\beta}{2}\int d\mathbf{r}\ \psi(\mathbf{r})V_Y^{-1}\psi(\mathbf{r}) + \beta\int d\mathbf{r}\ \rho(\mathbf{r})\psi(\mathbf{r})\right) \tag{4}$$

Let's substitute this back into our partition function Z. Notice how the spin density $\rho(\mathbf{r})$ now appears only in linear terms (to the power of 1). It couples to both the internal field $\psi(\mathbf{r})$ and the external field $h(\mathbf{r})$:

$$Z \propto \int \mathcal{D}\psi \exp\left(-\frac{\beta}{2}\int \psi V_Y^{-1}\psi\right) \sum_{\{s_i\}} \exp\left(\beta\int d\mathbf{r}\ \rho(\mathbf{r})\big[\psi(\mathbf{r}) + h(\mathbf{r})\big]\right) \tag{5}$$

1.3 Step 3: Returning to Discrete Using Delta Function

Now we encounter the most beautiful part of the derivation. We have a continuous integral containing $\rho(\mathbf{r})$. Let's substitute the original definition $\rho(\mathbf{r}) = \sum_i s_i\delta(\mathbf{r}-\mathbf{r}_i)$ back into the linear coupling term.

A fundamental property of the Dirac delta function is that integrating it with any function $f(\mathbf{r})$ simply evaluates that function at the spike's location: $\int f(\mathbf{r})\delta(\mathbf{r}-\mathbf{r}_i)d\mathbf{r} = f(\mathbf{r}_i)$. Therefore:

$$\beta\int d\mathbf{r}\left(\sum_i s_i\delta(\mathbf{r}-\mathbf{r}_i)\right)\big[\psi(\mathbf{r}) + h(\mathbf{r})\big] = \beta\sum_i s_i\big(\psi(\mathbf{r}_i) + h(\mathbf{r}_i)\big) \tag{6}$$

This rigorous step elegantly collapses the continuous integral back into a discrete sum over the lattice sites. The continuous fields ψ and h are perfectly "sampled" at the exact locations of the spins.

1.4 Step 4: Evaluating the Spin Trace (The Hyperbolic Cosine)

Because the spins are no longer multiplied by each other ($s_i \times s_j$), they are completely decoupled. We can calculate the sum over the states ($s_i = +1$ and $s_i = -1$) independently for each lattice site i:

$$\begin{aligned}\sum_{s_i=\pm1} \exp\left(\beta s_i(\psi_i + h_i)\right) &= \exp\left(+\beta(\psi_i + h_i)\right) + \exp\left(-\beta(\psi_i + h_i)\right)\\ &= 2\cosh\big(\beta(\psi_i + h_i)\big)\end{aligned} \tag{7}$$

Since this sum must be performed for every spin i in the lattice, the result is a massive product ($\prod$) over all sites:

$$\sum_{\{s_i\}} \exp(\ldots) = \prod_{i=1}^{N} 2\cosh\big(\beta(\psi_i + h_i)\big) \tag{8}$$

1.5 Step 5: The Continuum Limit via Riemann Sum

To complete our field theory, we want to convert this discrete product back into a continuous spatial integral. We start by taking the logarithm, which turns the product into a sum: $\sum_{i=1}^{N} \ln\big[2\cosh(\ldots)\big]$.

Let $\Delta V = a^D$ be the volume of a single D-dimensional unit cell. We can multiply and divide the sum by this volume, structuring it exactly like a Riemann sum from basic calculus:

$$\sum_{i=1}^{N} \Delta V \left[\frac{1}{a^D} \ln\big[2\cosh(\ldots)\big]\right] \tag{9}$$

The Long-Wavelength Assumption: If we assume that the lattice spacing a is extremely small, and that the fields ψ and h change very smoothly over space, we can take the continuum limit. The discrete sum of volume elements seamlessly transforms into a spatial integral ($\sum \Delta V \to \int d\mathbf{r}$):

$$\int d\mathbf{r}\ \frac{1}{a^D} \ln\Big[2\cosh\big(\beta(\psi(\mathbf{r}) + h(\mathbf{r}))\big)\Big] \tag{10}$$

2 The Master Equation: Non-linear Ginzburg-Landau Theory

In this report, we analyze the effect of a local external field $h(\mathbf{r})$ on the order parameter distribution of a system, based on the exact non-linear Ginzburg-Landau effective free energy functional derived via the Hubbard-Stratonovich (HS) transformation. The effective free energy density is given by:

$$\beta\mathcal{F}[\psi] = \int d\mathbf{r} \left\{\frac{\beta}{2}\psi(\mathbf{r})V_Y^{-1}\psi(\mathbf{r}) - \frac{1}{a^D} \ln\Big[2\cosh\big(\beta(\psi(\mathbf{r}) + h(\mathbf{r}))\big)\Big]\right\} \tag{11}$$

To find the equilibrium state of the natural system, we apply the calculus of variations by setting the functional derivative to zero ($\delta\mathcal{F}/\delta\psi = 0$). Since the derivative of $\ln(\cosh x)$ is $\tanh x$, we obtain the following exact, non-linear Euler-Lagrange differential equation, which serves as our Master Equation:

$$c(-\nabla^2 + \kappa^2)\psi(\mathbf{r}) - \frac{1}{a^D} \tanh\big[\beta(\psi(\mathbf{r}) + h(\mathbf{r}))\big] = 0 \tag{12}$$

This equation is the rigorous governing equation of the system, containing no mathematical approximations.

3 Ginzburg-Landau Expansion and Coefficient Mapping

To compare the physical implications of the Master Equation with standard phase transition theory, we perform a Taylor expansion of the non-linear term in the free energy near the critical point (where the fluctuation field ψ is small). Defining the inverse of the lattice volume

element as the spin density $\rho_0 \equiv 1/a^D$, and substituting $x = \beta(\psi + h)$ into the Maclaurin series $\ln(2\cosh x) \approx \ln 2 + \frac{1}{2}x^2 - \frac{1}{12}x^4$, we obtain the inhomogeneous Ginzburg-Landau form:

$$\mathcal{F}_{GL}[\psi] = \int d\mathbf{r} \left[\frac{C}{2}(\nabla\psi)^2 + \frac{A(\mathbf{r})}{2}\psi^2 + \frac{U}{4}\psi^4 - H_{\text{eff}}(\mathbf{r})\psi \right] + F_0 \tag{13}$$

Mapping each GL coefficient to the microscopic parameters derived from the HS transformation yields the following:

- **Gradient Coefficient (C):** $C = c$
- **Quartic Coefficient (U):** $U = \frac{\rho_0\beta^3}{3}$
- **Effective Field (H_{eff}):** $H_{\text{eff}}(\mathbf{r}) = \rho_0\beta h(\mathbf{r}) - \frac{\rho_0\beta^3}{3}h^3(\mathbf{r})$
- **Local Mass/Temperature Coefficient (A):**

$$A(\mathbf{r}) = c\kappa^2 - \rho_0\beta + \rho_0\beta^3 h^2(\mathbf{r}) \tag{14}$$

In the standard GL theory, the external field is treated merely as a linear perturbation ($-h\psi$). However, in our derivation, $h(\mathbf{r})$ resides inside the cosh function, inducing non-linear coupling terms ($6h^2\psi^2$) upon expansion. The most crucial observation is the addition of the $+h^2(\mathbf{r})$ term to the quadratic coefficient $A(\mathbf{r})$. This implies that the phase transition condition ($A = 0$) changes in localized regions where $h(\mathbf{r})$ exists. Consequently, this mathematically proves that the external field not only breaks symmetry but directly shifts the **local critical temperature (Local T_c)**.

4 Linear Approximation and Geometric Case Studies

In the small-signal regime where the temperature is above the critical point ($T \gtrsim T_c$) and the external field h_0 is weak, the fluctuation field ψ is small. Thus, we can perform a linearization by ignoring higher-order terms (ψ^3, h^2, h^3). Accordingly, the local coefficient $A(\mathbf{r})$ is approximated by the bulk constant $A_0 = c\kappa^2 - \rho_0\beta$, and the governing equation simplifies to the following linear inhomogeneous differential equation:

$$C\nabla^2\psi(\mathbf{r}) - A_0\psi(\mathbf{r}) = -\rho_0\beta h(\mathbf{r}) \tag{15}$$

We define the correlation length as $\xi = \sqrt{C/A_0}$. We now derive the spatial profiles for two specific geometric structures under a local external field.

4.1 Case A: Semi-infinite System ($x \geq 0$)

Assume a semi-infinite system where the external field h_0 acts only at the surface ($x = 0$) and vanishes deep within the bulk ($x \to \infty$).

- **Governing Equation:** For $x > 0$, the external field is zero, yielding a homogeneous equation:

$$\frac{d^2\psi}{dx^2} - \frac{1}{\xi^2}\psi = 0 \tag{16}$$

- **General Solution:** The standard solution for this second-order linear ODE is:

$$\psi(x) = C_1 \exp\left(-\frac{x}{\xi}\right) + C_2 \exp\left(\frac{x}{\xi}\right) \tag{17}$$

- **Applying Boundary Conditions:** First, the physical requirement that the order parameter must not diverge in the bulk imposes $\psi(\infty) = 0$. Therefore, we must set $C_2 = 0$.

$$\psi(x) = C_1 \exp\left(-\frac{x}{\xi}\right) \tag{18}$$

  Second, at the surface ($x = 0$), the strong external field induces a fixed boundary order parameter $\psi(0) = \psi_s$ (where $\psi_s \propto h_0$). Substituting $x = 0$ gives $C_1 = \psi_s$.

- **Final Solution:**

$$\psi(x) = \psi_s \exp\left(-\frac{x}{\xi}\right) \tag{19}$$

Physically, this indicates that the order induced at the surface penetrates the bulk with an exponential decay characterized by the correlation length ξ. As the temperature approaches the critical point ($A_0 \to 0$), $\xi \to \infty$, causing the surface order to spread throughout the entire bulk, triggering a Critical Wetting phenomenon.

4.2 Case B: Periodic Ring System (L)

Consider a 1D periodic ring of circumference L, where a delta-function external field $h(x) = h_0\delta(x)$ is applied exclusively at the origin ($x = 0$).

- **Governing Equation:**

$$C\frac{d^2\psi}{dx^2} - A_0\psi = -\rho_0\beta h_0\delta(x) \tag{20}$$

- **General Solution Formulation:** For $x \neq 0$, the equation is homogeneous ($C\psi'' - A_0\psi = 0$). Due to the periodic boundary condition $\psi(x) = \psi(x+L)$ and the structural symmetry around the origin $\psi(x) = \psi(-x)$, the solution in the interval $x \in [-L/2, L/2]$ must be an even function centered at $x = L/2$. Thus, we construct the solution using the hyperbolic cosine:

$$\psi(x) = D\cosh\left(\frac{L/2 - |x|}{\xi}\right) \tag{21}$$

- **Determining the Constant D via Delta Function Singularity:** To find the constant D, we integrate the governing differential equation across the infinitesimally small interval $[-\epsilon, \epsilon]$ around $x = 0$:

$$\int_{-\epsilon}^{\epsilon}\left(C\frac{d^2\psi}{dx^2} - A_0\psi\right)dx = \int_{-\epsilon}^{\epsilon} -\rho_0\beta h_0\delta(x)dx \tag{22}$$

  Since $\psi(x)$ is continuous, the integral of $A_0\psi$ vanishes as $\epsilon \to 0$. The integral of the second derivative yields the jump in the first derivative:

$$C\left[\left.\frac{d\psi}{dx}\right|_{0^+} - \left.\frac{d\psi}{dx}\right|_{0^-}\right] = -\rho_0\beta h_0 \tag{23}$$

  Using our assumed solution for $x > 0$, $\psi(x) = D\cosh((L/2 - x)/\xi)$, the derivative is:

$$\psi'(x) = -\frac{D}{\xi}\sinh\left(\frac{L/2 - x}{\xi}\right) \quad \Longrightarrow \quad \psi'(0^+) = -\frac{D}{\xi}\sinh\left(\frac{L}{2\xi}\right) \tag{24}$$

By symmetry, $\psi'(0^-) = -\psi'(0^+)$. Substituting these into the jump condition:

$$\begin{aligned} C\left[-\frac{D}{\xi}\sinh\left(\frac{L}{2\xi}\right) - \frac{D}{\xi}\sinh\left(\frac{L}{2\xi}\right)\right] &= -\rho_0\beta h_0 \\ -\frac{2CD}{\xi}\sinh\left(\frac{L}{2\xi}\right) &= -\rho_0\beta h_0 \\ D &= \frac{\rho_0\beta h_0\xi}{2C\sinh(L/2\xi)} \end{aligned} \tag{25}$$

- **Final Solution:**

$$\psi(x) = \frac{\rho_0\beta h_0\xi}{2C\sinh(L/2\xi)}\cosh\left(\frac{L/2 - |x|}{\xi}\right) \tag{26}$$

In a closed ring structure, the order induced by the local field propagates in both directions and creates constructive interference at the opposite end ($x = L/2$), forming a smooth U-shaped cosh profile.

5 Critical Phenomena and Finite-Size Scaling

5.1 Bulk Critical Temperature (T_c)

The intrinsic critical temperature of the system (Bulk T_c) is determined at the point where the bulk coefficient A_0 becomes zero in the absence of an external field, causing the correlation length to diverge.

$$T_c = \frac{\rho_0}{k_B c\kappa^2} \tag{27}$$

The critical temperature is proportional to the spin density ρ_0 and inversely proportional to the square of the Yukawa screening parameter κ. As the interaction range shortens (κ increases), it becomes harder to overcome thermal fluctuations, resulting in a sharp drop in T_c.

5.2 Finite-Size Scaling in Periodic Ring (Case B)

While a true macroscopic phase transition does not occur in a ring of finite size L, a crossover takes place under the condition $\xi \approx L$, where the entire system falls under the dominance of the local field h_0. The pseudo-critical temperature T^* at this point follows a Finite-Size Scaling relation:

$$T^* \approx T_c\left(1 + \frac{C}{c\kappa^2 L^2}\right) \tag{28}$$

Mathematically, the smaller the system size ($L \to 0$), the more maximized the interference effect of the local field becomes. Consequently, the formation of order initiates at a temperature higher than the bulk T_c ($\Delta T \propto 1/L^2$).

6 Proposed Verification via Monte Carlo Simulation

To validate these theoretical derivations and extend the study into the non-linear regime, we propose a Metropolis Monte Carlo simulation. Since simulations do not rely on mathematical approximations, they serve as a powerful tool to cross-validate the non-linearity of the Master Equation.

1. **Linear vs Non-linear Regime:** We aim to demonstrate that the spin profile $\langle s_i \rangle$ obtained from the simulation perfectly matches the theoretical linear solutions (exp and cosh) in the $T \gtrsim T_c$ regime. Conversely, when the temperature drops significantly or h_0 becomes strong enough to break the linear approximation, we expect to observe kink-like penetration profiles caused by the non-linear terms of the Master Equation.

2. **Scaling Validation:** By varying the ring size L, we will statistically verify whether the shift in the extracted T^* follows the theoretically predicted $1/L^2$ trajectory.

3. **Screening Effect:** We will track the collapse of the correlation length ξ as the Yukawa screening parameter κ changes, thereby confirming the critical universality with the short-range Ising model.

7 Conclusion

In this study, we applied the Hubbard-Stratonovich transformation to the microscopic Ising-Yukawa model to derive a rigorous non-linear field-theoretic Master Equation. Through Taylor expansion, we mathematically proved the non-linear coupling mechanism by which an external field $h(\mathbf{r})$ modulates the local critical temperature. Furthermore, using linearized governing equations, we successfully elucidated the propagation profiles of order (exp and cosh forms) under specific geometric conditions, as well as the finite-size scaling law ($\Delta T \propto 1/L^2$). This framework provides a comprehensive theoretical foundation for understanding surface critical phenomena and phase transitions in nano-confined spaces within Soft Matter systems, and will serve as a crucial benchmark for future non-linear simulation studies and Electrical Double Layer (EDL) dynamics analysis.